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Setting Up Active Directory in Oracle VirtualBox: A Comprehensive Home Lab Guide

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Introduction:

Setting up an Active Directory (AD) environment within the confines of your home lab using Oracle VirtualBox presents an excellent opportunity to immerse yourself in the intricacies of network administration and Windows server technologies. Throughout this comprehensive guide, I will meticulously outline each step involved in configuring a virtualized network environment, encompassing a domain controller, DHCP server, and client machines.

Overview:

This tutorial serves as a detailed walkthrough for establishing a home lab environment utilizing Oracle VirtualBox. Our primary objective is to construct a virtualized network environment comprising essential components such as a domain controller equipped with Active Directory, a DHCP server, and client machines. Below, I provide an organized overview of the sequential steps we'll undertake:

1. **Download and Install Oracle VirtualBox**: We'll begin by acquiring and installing Oracle VirtualBox, laying the foundation for our virtualized network infrastructure.

2. **Obtain Windows 10 and Windows Server 2022 ISO Files**: We'll procure the necessary ISO files for Windows 10 and Windows Server 2022, essential for subsequent installations within our virtual environment.

3. **Create the Domain Controller Virtual Machine**: This step involves configuring the domain controller virtual machine, a pivotal element in our network architecture.

4. **Install Windows Server 2022 and Configure IP Addressing**: We'll proceed to install Windows Server 2022 on the domain controller VM and establish requisite IP addressing configurations.

5. **Set Up Active Directory and DNS Services**: Here, we'll configure Active Directory and DNS services on the domain controller, pivotal for network operations.

6. **Configure Routing for Internet Access**: Facilitating internet access from the private network requires configuring routing protocols, a crucial aspect we'll address comprehensively.

7. **Set Up DHCP Server**: Automatic IP address assignment to client machines necessitates configuring the DHCP server on the domain controller.

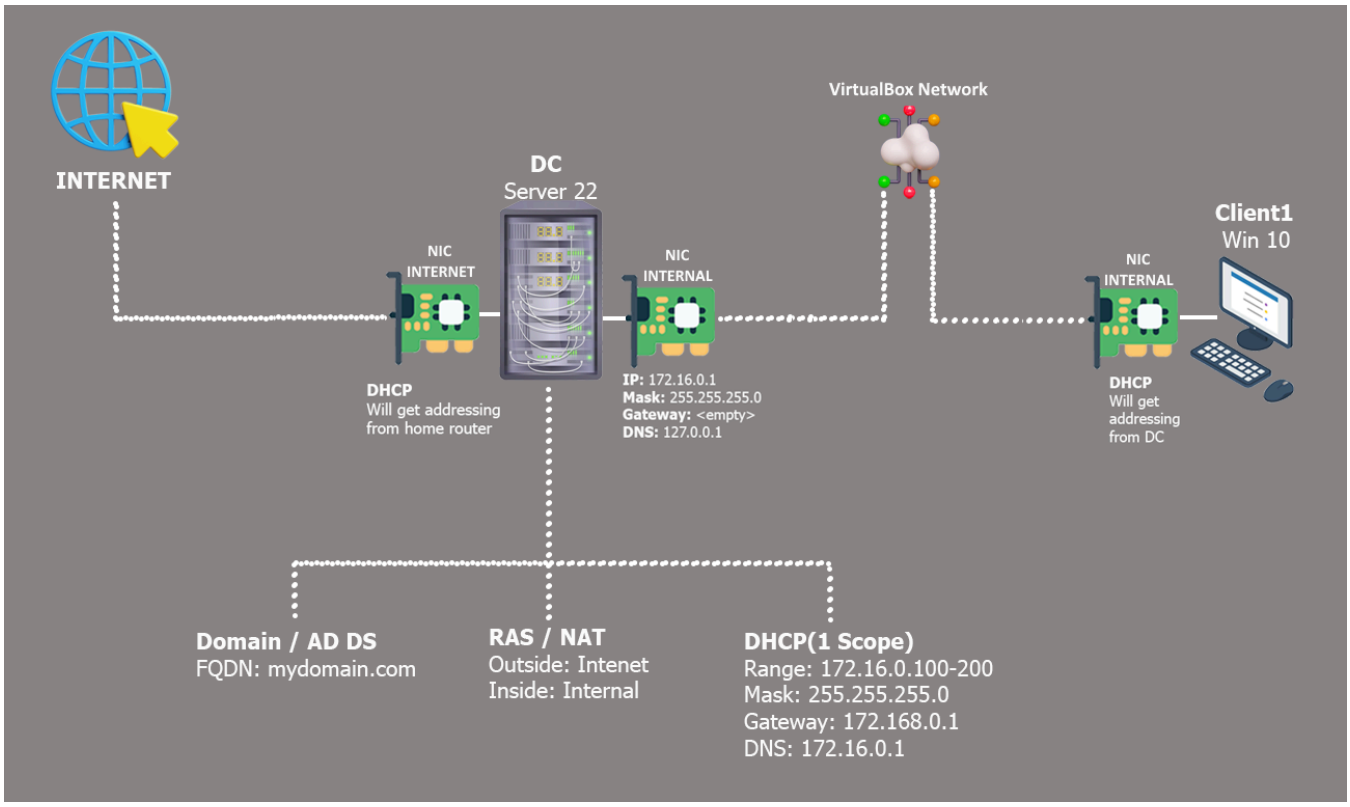
8. **Run PowerShell Script for User Creation**: We'll employ PowerShell scripting to automate user creation within Active Directory, streamlining administrative tasks.

9. **Create Client Virtual Machine and Install Windows 10**: This step involves creating a client virtual machine and installing Windows 10, pivotal for domain integration.

10. **Join Client Machine to the Domain**: Lastly, we'll integrate the client machine into the domain and authenticate user logins using domain credentials.

Visualization:

To help visualize the process, refer to the following diagram:



Network Diagram

Lab Requirements

- **Virtualization Software:** *Oracle VirtualBox and Oracle VM VirtualBox Extension Pack (options can be VMware Player)*
- **Windows Server 2022**. The evaluation copy can be found on the Microsoft Evaluation Centre - Link
- **Windows 10** - Link

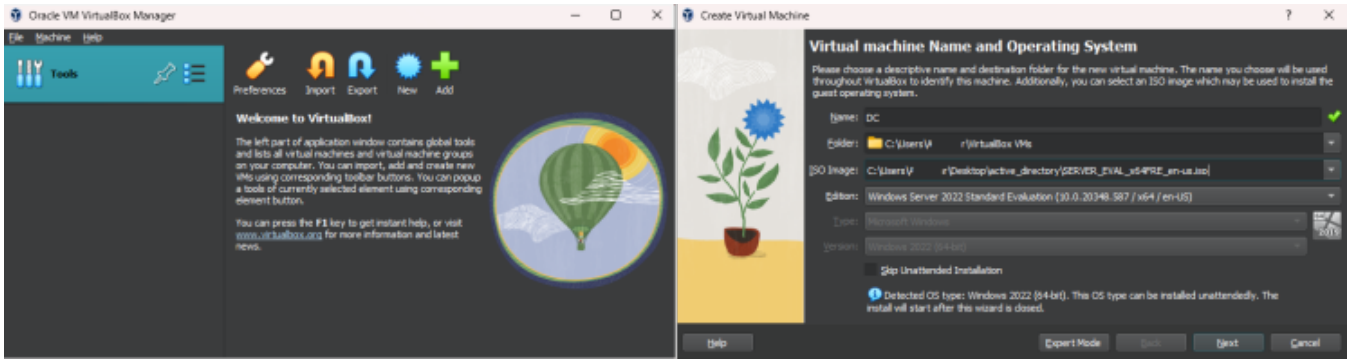
💡 *Virtual Box and VMware Player are free to download.*

Now, without further ado, let's get started.

Windows Server 2022 Setup

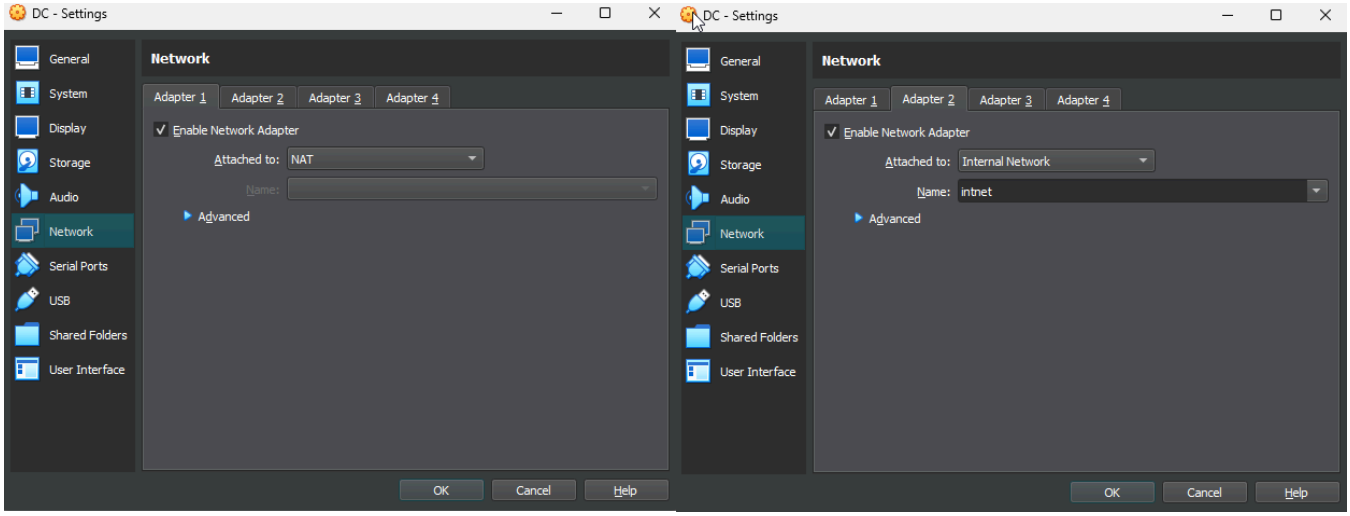
Once the ISO file is obtained from the Microsoft Evaluation Centre, the next step involves setting up a new virtual machine. I am using Oracle VirtualBox for the lab.

1. To create a new virtual machine, begin by clicking on Oracle VirtualBox **"New"**. Then, access the drop-down menu next to **"ISO Image"** and select the **"Windows Server ISO file"** you have downloaded. Additionally, you can choose the "Skip Unattended Installation" option to speed-up the setup process.



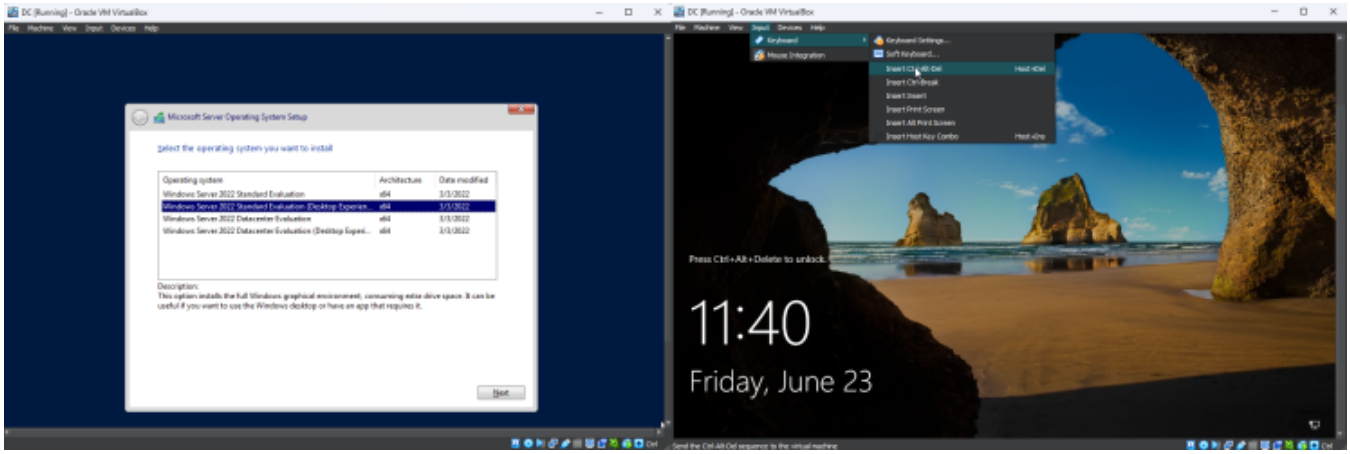
2. After completing the VM initialization setup and clicking on the "Finish" button, we will configure the network settings with two network adapters. One adapter will be connected to the external internet, while the other will be linked to the private lab network.

- Adapter #1 — NAT: Connects to the internet.
- Adapter #2 — intnet: Connects to the private lab network.



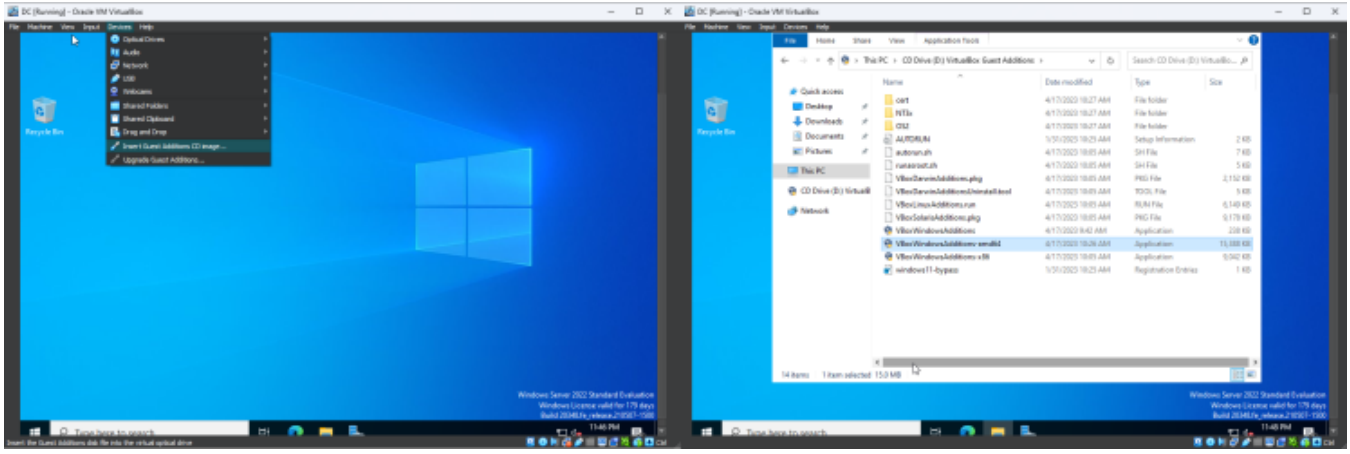
VM initialization setup.

3. During VM initialization setup, it's important to note that the conventional method of pressing **Ctrl+Alt+Delete** may not operate as anticipated. To unlock the system, you have two options: either navigate to **"Input"** then **"Keyboard"** and select **"Insert Ctrl+Alt+Del"**, or use the keyboard shortcut **"Ctrl (left) + Del"**.



Finalizing installion

4. After logging into the server, we'll configure before installing Active Directory Services. First, we'll install VM Guest Additions to enhance the user experience, fixing issues like mouse lag and screen resizing. Navigate to **"Devices" > "Insert Guest Additions CD Image..."**, locate "**VBoxWindowsAdditions-amd64"** in the CD Drive, run the file, follow prompts to install, and reboot the system.

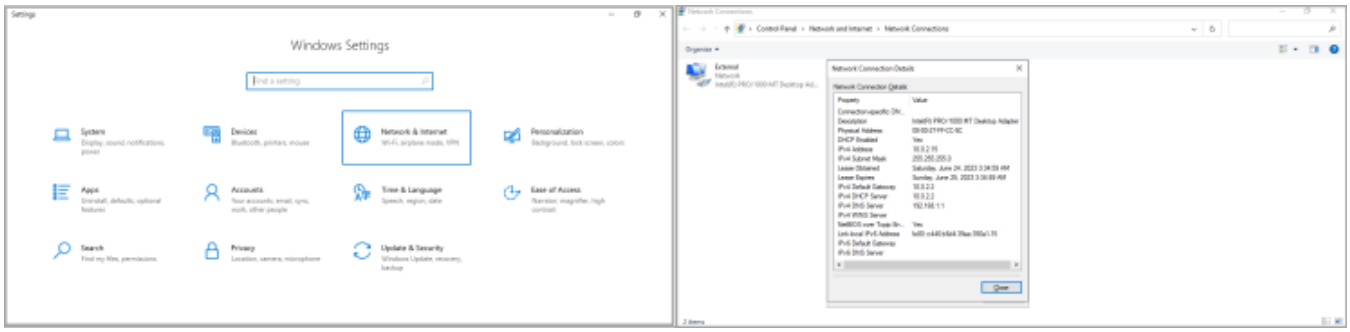


Installing VM Guest Additions

5. In the following step, we will proceed with configuring the two network adapters that were added during the initial setup of the virtual machine. To begin, navigate to the **"Settings"** page, then click on the **"Network and Internet"** section, and select **"Change Adapter Options."**

For the first adapter, labelled as **"Ethernet0,"** we will keep it unchanged on the **NAT** network, allowing it to acquire an IP address from the home router. In my case, it is assigned to the **10.0.2.15/24** subnet.

We will rename the adapters one to **External** and the other to **Internal** to distinguish between the networks.



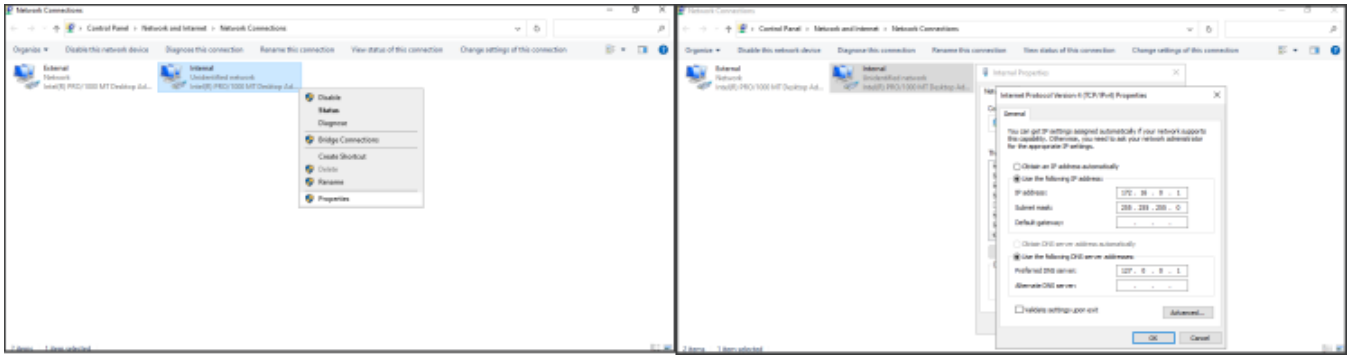
Renaming the NAT network to External (internet).

6. To configure the Internal network adapter, I have chosen the **172.16.0.0/24** range. Consequently, I have assigned the server an IP address of **172.16.0.1** and a subnet mask of **"255.255.255.0"**.

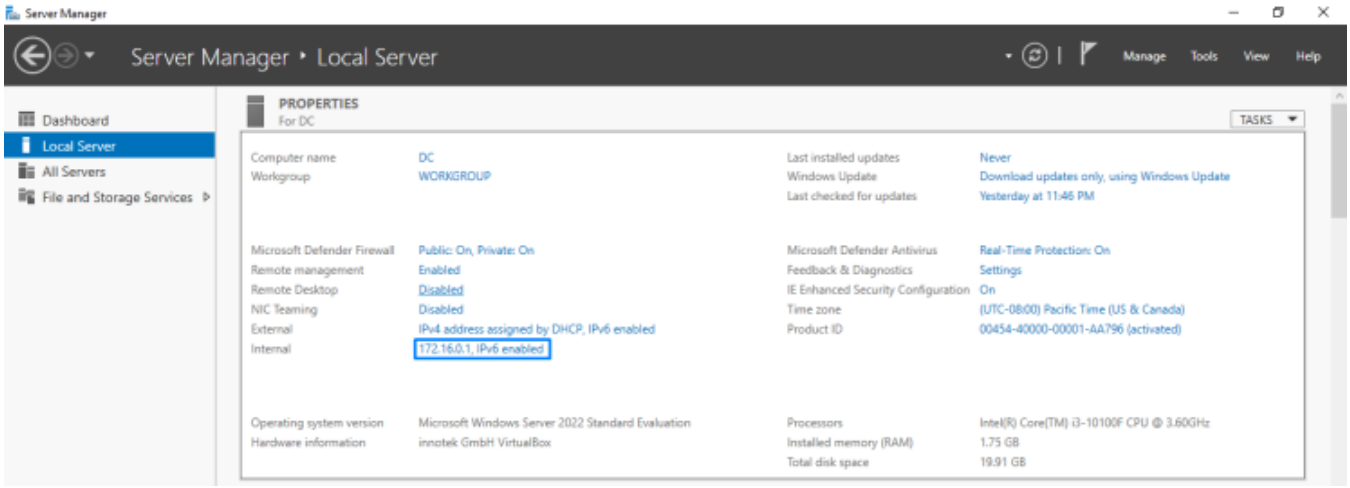
As the server IP will eventually become the domain controller after installing Active Directory services and promoting it, the Default Gateway field remains empty. This

is because the Domain Controller itself will act as the gateway.

For the DNS, we assign it the DC address, which is **172.16.0.1**, or a loopback of **127.0.0.1** so it pings itself. Both ways are correct.



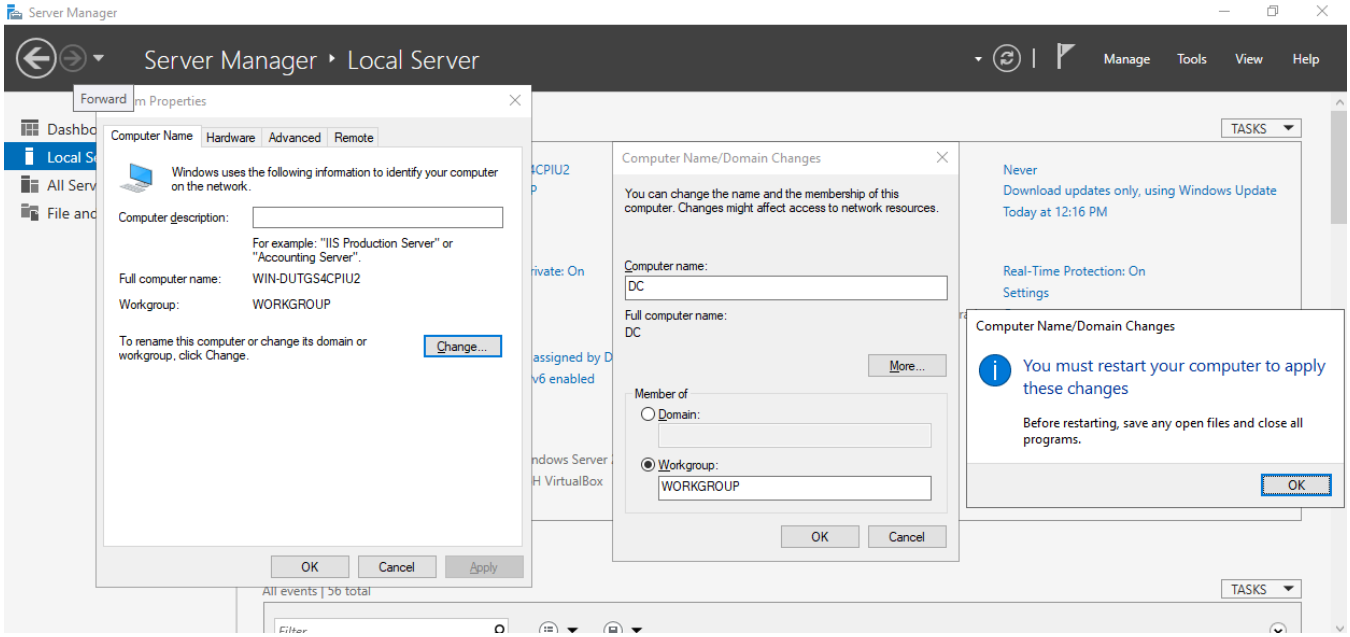
Once you have finished configuring the IP settings, navigate to the **"Server Manager"** and update the **"Local Server"** dashboard. By doing so, you will be able to view the IP address assigned to the server.



Internal network IP is assigned to 172.16.0.1

7. The last step is to rename the server to something meaningful; since it will be the domain controller in the domain. I called it **"DC"**

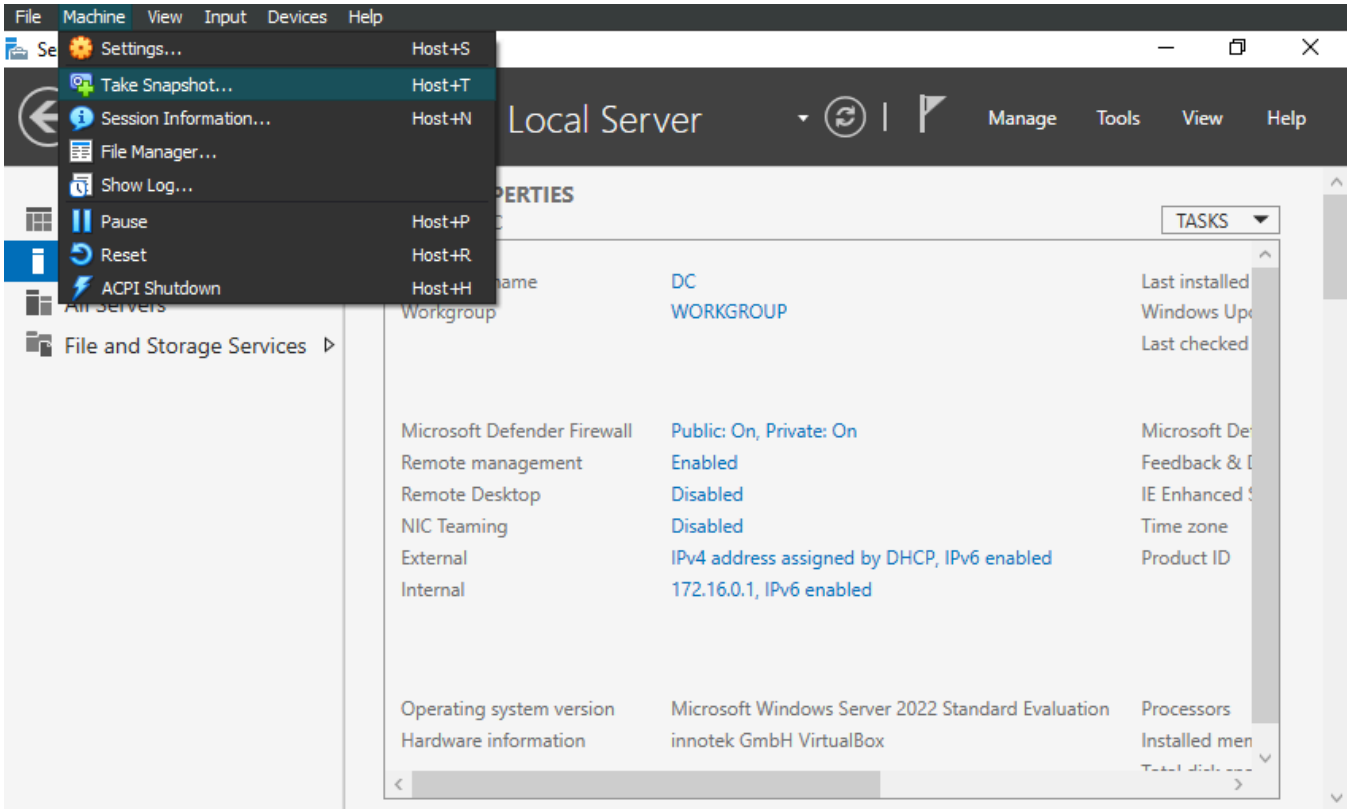
And restart the server to apply changes.



Rename the server.

After restarting the server, log in to it and verify that the changes have been successfully implemented. Then, create a snapshot of the machine with the updated configuration.

🚨 *Having a snapshot of the current VM state before installing Active Directory is crucial. In the event that you encounter any issues during the process and make a mistake, you can easily revert back to the initial state and start again from there.*



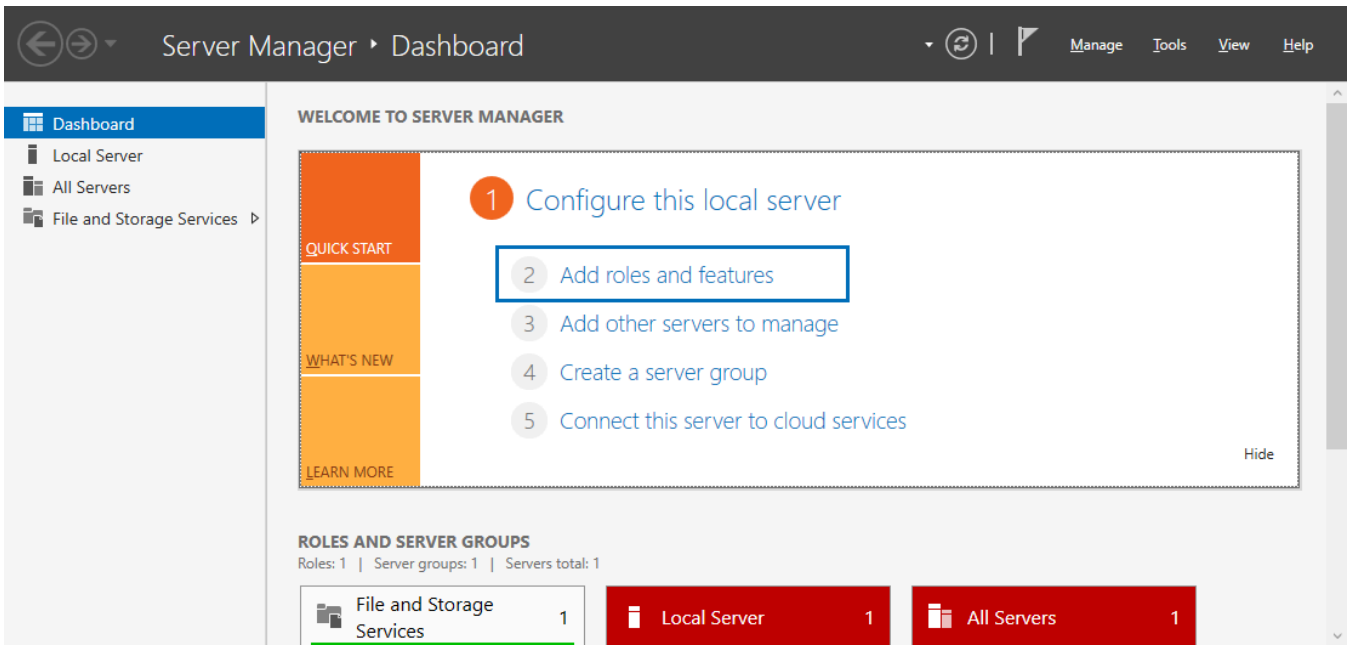
Steps to take a Snapshot of the current VM state.

Configuring Active Directory Services

Active Directory Setup

- To get started with installing Active Directory services and creating the forest root domain called **"mydomain.com"**, we click on the **"Add roles and features"** option in the server's dashboard. This enables us to add the AD role and begin the setup process.

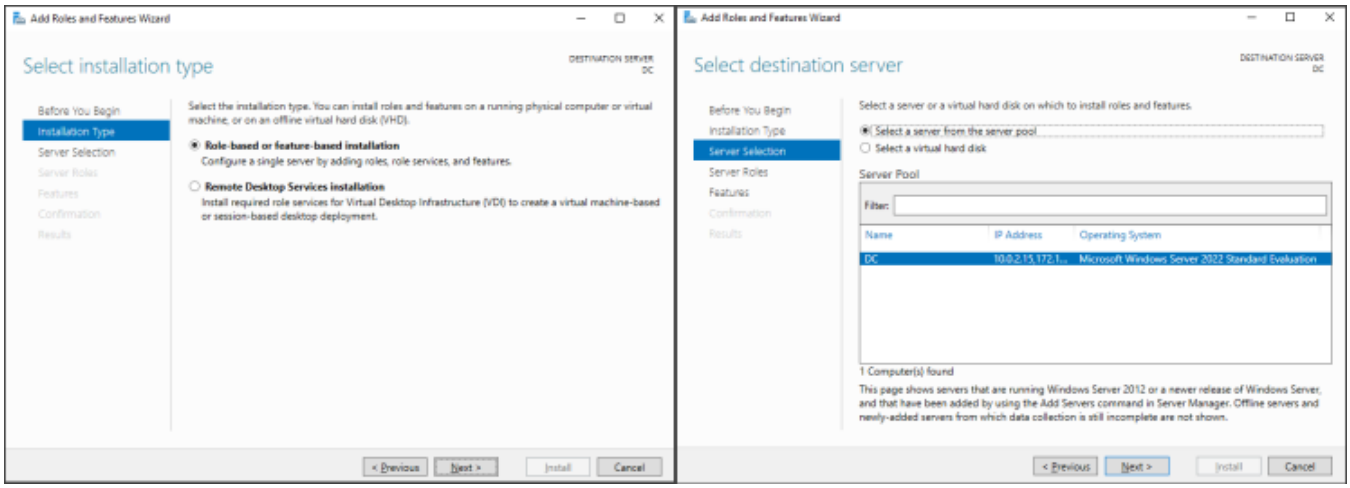
💡 *When it comes to the server's role within a network, there's a difference between "roles" and "features." In this lab, the role refers to the server's specific function, which is acting as a domain controller with the Active Directory role. On the other hand, features are the extra capabilities that come along with the role, like AD administration tools, AD DS tools, Group Policy Management, and so on.*



Adding new roles and Features.

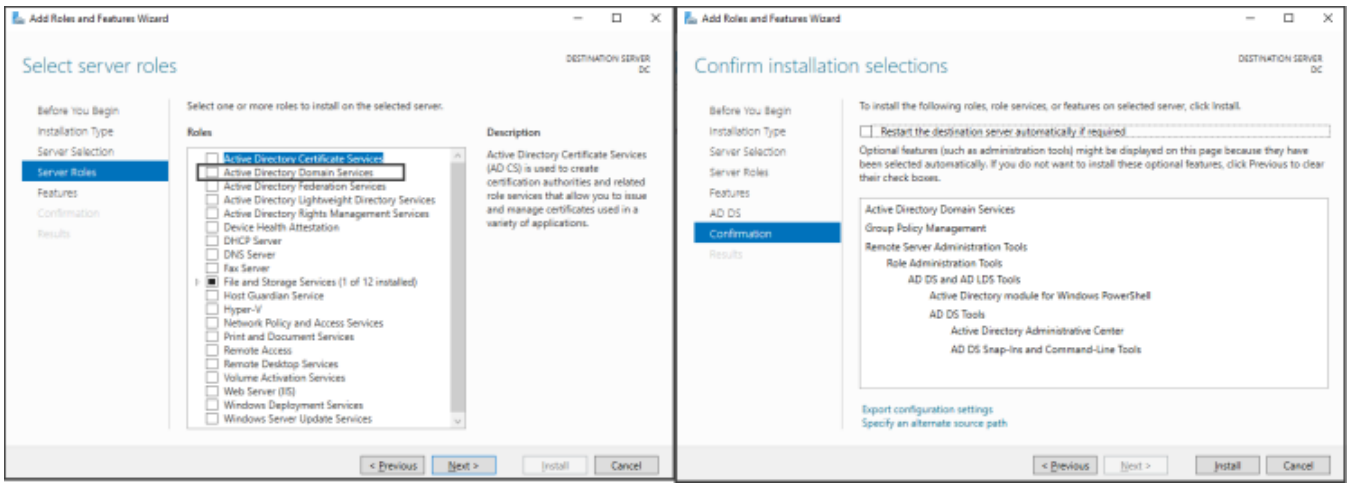
- Within the Setup Wizard, opt for the installation option labeled as "Role-based or feature-based." This choice is appropriate because the role will be added directly to the virtual machine, rather than being installed remotely via a Virtual Desktop Infrastructure (VDI).

- Proceed by choosing the server "**DC0**" from the list of available servers in the Server Pool. Once selected, click on the "**Next**" button to continue.



Adding the role-based installation and selecting the Domain Controller (DC0) from the servers list.

- In the Server Roles section, select **'Active Directory Domain Services'** and click **'Add Features.'** Then, proceed through the installation wizard without modifying anything in the **'Features'** and **'AD DS'** sections. Click **'Next'** until you reach the **'Confirmation'** section, where you'll click **'Install'** to begin the installation process.



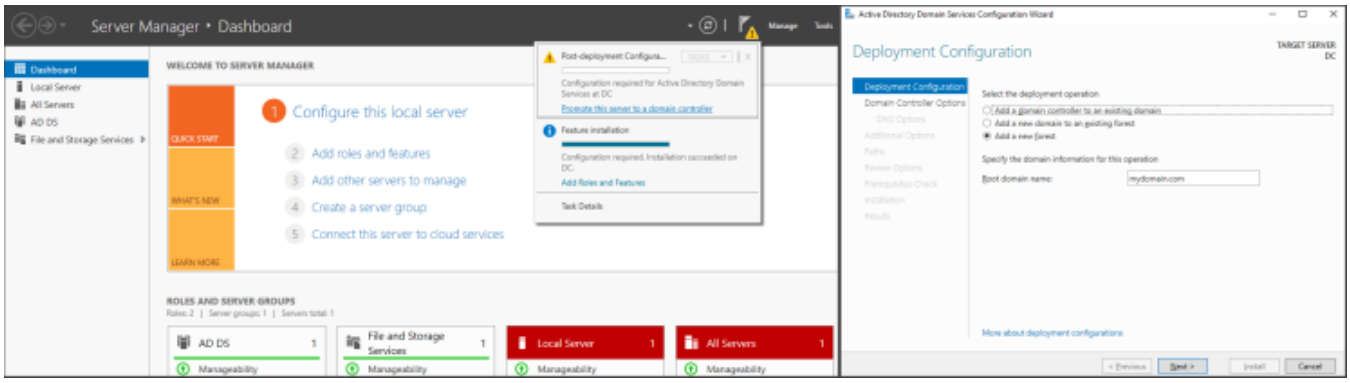
Selecting the AD DS role and Confirmation sections.

- Once the installation process is finished, a notification will appear regarding the Post-deployment Configuration. This notification will inquire whether you wish to promote the server to a domain controller.

- Click on the notification to initiate the Domain Controller configuration. From there, choose the option to **"Add a new forest"** and assign the name **"mydomain.com"** to the lab forest.

💡 *A forest is a big container that holds all objects and configurations in the Active Directory structure (ex: child domains, users, computers, groups, policies, schemas, etc.)-*

- Additionally, a **forest serves as a security boundary**, signifying that objects within separate Active Directory forests cannot interact with one another unless a trust relationship is established between the desired forests.



Promoting the server to a domain controller and creating a new forest.

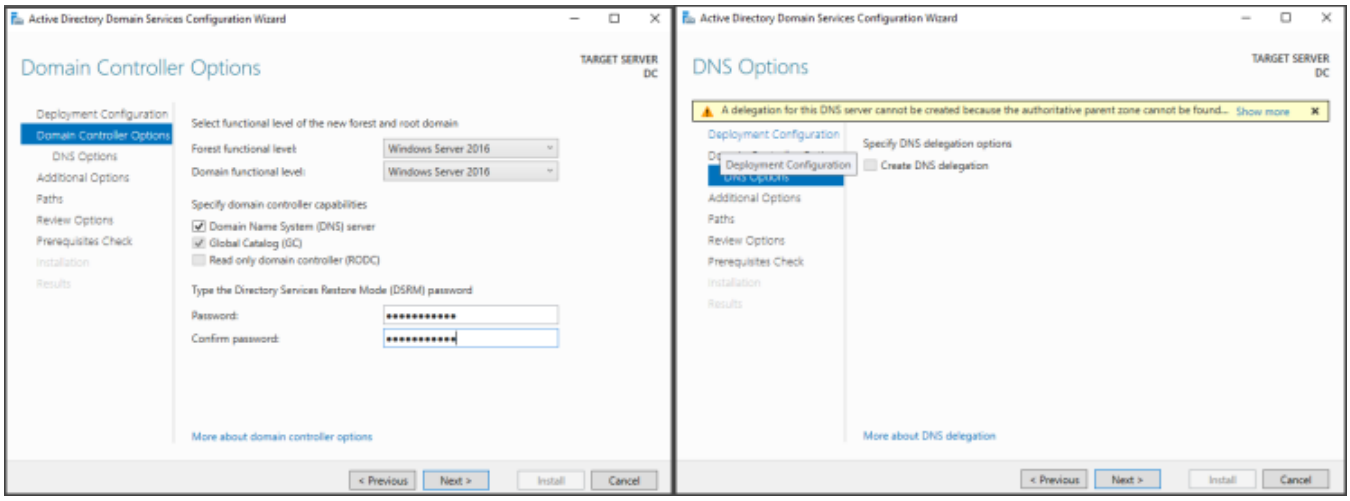
- Next, we verify that the **"Functional Forest level"** is set to Windows 2016, which represents the highest level accessible, and select a password for the **"Directory Services Restore Mode (DSRM)."**Regarding the **"DNS Options,"** we will leave the **"Create DNS Delegation"** option unchecked and proceed by clicking "Next."

💡*The functional level is the capabilities that Active Directory supports, and it also determines which Windows Server operating systems you can run on domain*

controllers in the forest. ***The Windows 2016 functional level supports running Windows 2022 on the DC.***

💡***Directory Services Restore Mode (DSRM)*** *is a special boot mode only available on the domain controller that allows the domain administrators to log into the domain controller using the DSRM password when the Active Directory fails. It is the local administrator account for the domain controller server.*

💡***Safe mode grants Domain Administrators (DAs)*** *access to the Active Directory database for the purpose of restoring or repairing objects.*

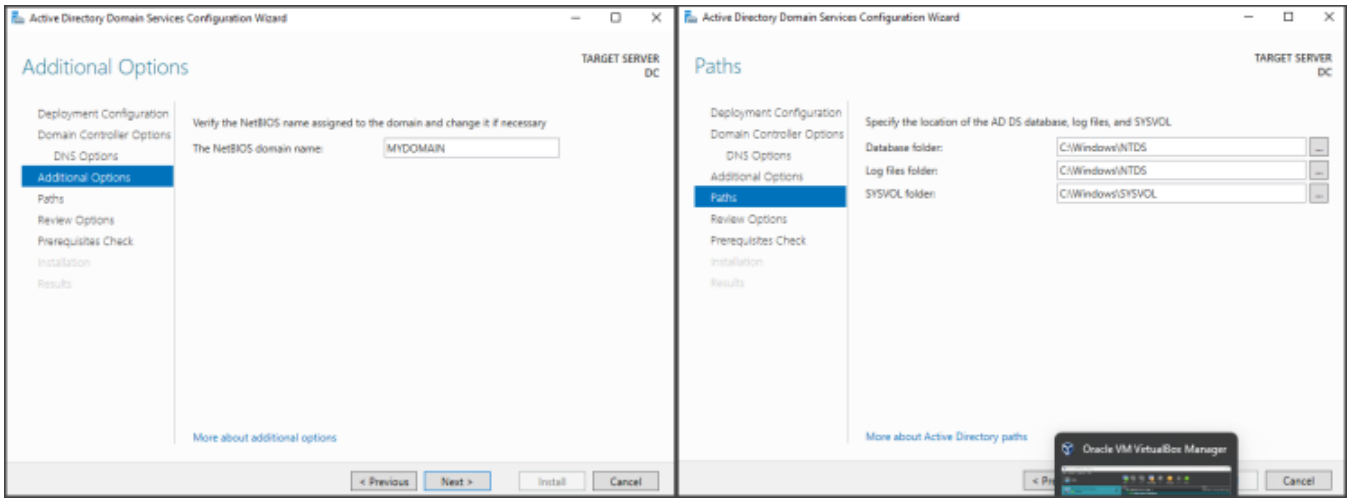


Setting up the DSRM password for the domain controller and DNS Options.

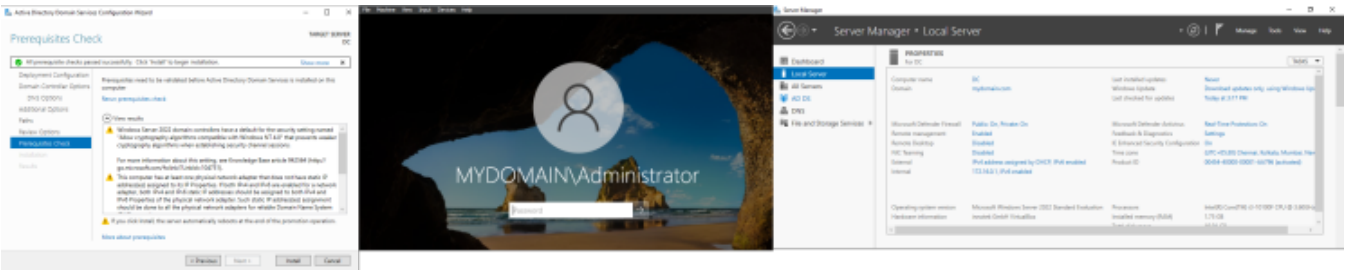
- Verify the **NetBIOS** domain name and click “Next” to move to the **“Paths”** section.The path locations specified are for the Active Directory database (NTDS), the Log files, and the SYSVOL folder. It is advisable to retain the default paths to avoid any potential issues in the future. We will proceed by clicking **"Next."**

💡 ***NTDS*** *is a database that stores Active Directory data, including information about users, computers, groups, and network resource objects.*

💡 ***SYSVOL*** *folder is situated directly on the domain controller, housing various public files, folders, and essential components like Group Policy Objects (GPOs) and scripts. These resources are utilized for effective management of domain users and computers within the forest.*



- Now that we have everything set up for the Active Directory, we can get started with the installation process by simply clicking on the **"Install"** button in the **"Pre-requisite Check"** section. It'll only take a few seconds, and once it's done, you'll need to reboot your machine.Once the machine has been restarted, we can proceed to the server dashboard and verify the presence of the newly created domain **"mydomain.com"**.



Installing the AD DS, rebooting the server, and verifying the newly created domain.

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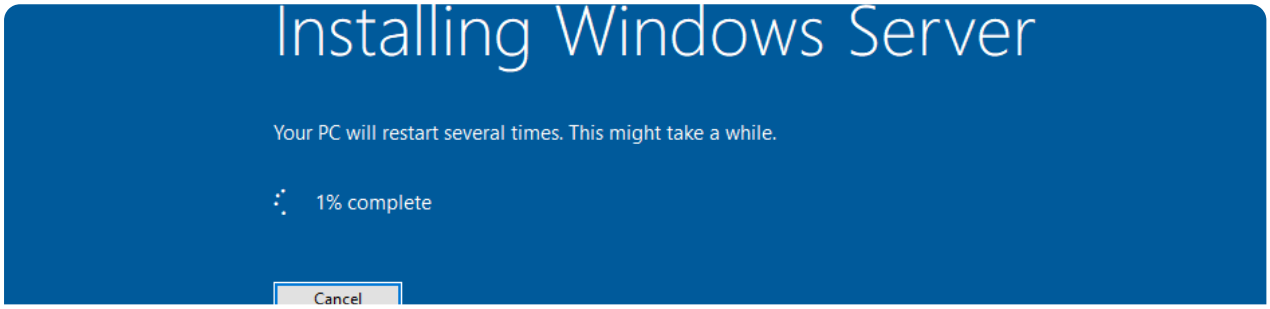

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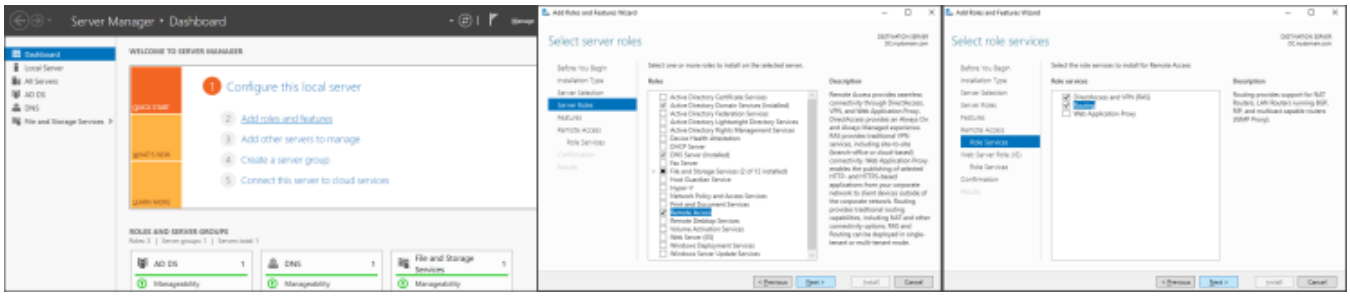
Installing RAS/NAT

- We will install **Remote Access Server (RAS)** and **Network Address Translation (NAT)** on the domain controller. This installation aims to enable our **Windows 10 client**, which we will set up later, to connect to a private virtual network while still having internet access through the domain controller.

- This configuration ensures that our clients can effectively access the internet while connected to the private network using the RAS and NAT components installed on the domain controller.

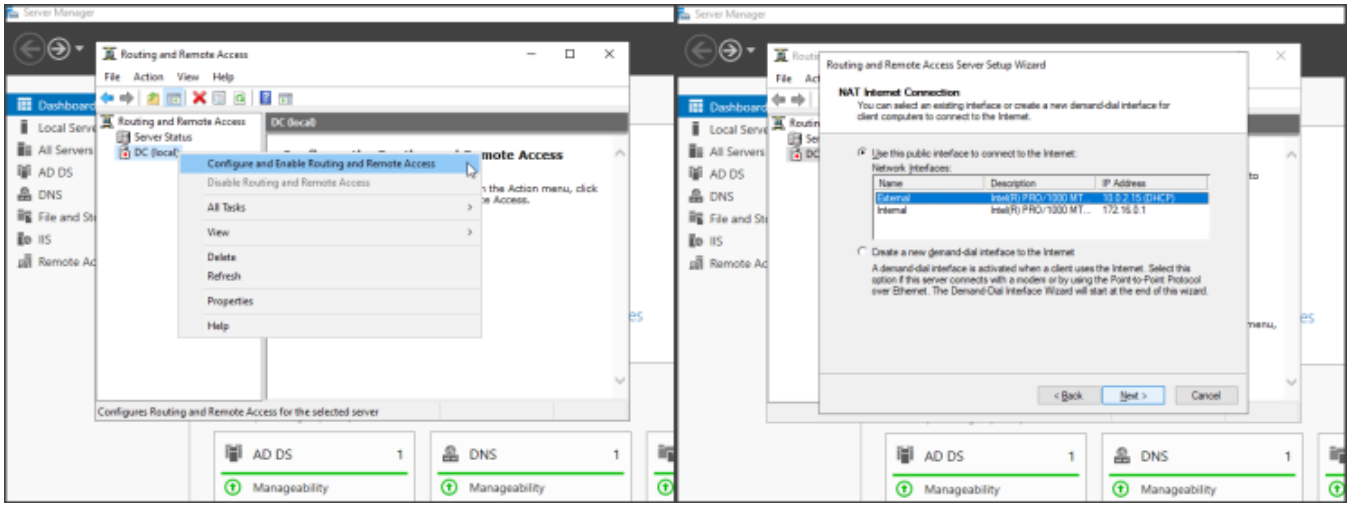
- Launch "Server Manager" and select "Add new roles and features." Click 'Next' until reaching the "Server Roles" section. Choose "Remote Access" as the role. Proceed by clicking 'Next' to "Role Services," select "Routing," and click "Add features." Move to the "Confirmation" section, and click "Install" to begin the installation process.



Adding new roles and features, selecting Server Roles and Routing option.

To complete the setup, open **"Server Manager"** and select **"Tools" -> "Routing and Remote Access."** Right-click on your domain controller **(e.g., "DC")** and choose **"Configure and Enable Routing and Remote Access."**

- Follow the Setup wizard by selecting **"Next"** and choose **"Network Address Translation (NAT)."** In the NAT Internal Connection window, select the connection with internet access (e.g., "External"). Click **"Next"** and then **"Finish"** to successfully install **RAS/NAT.**



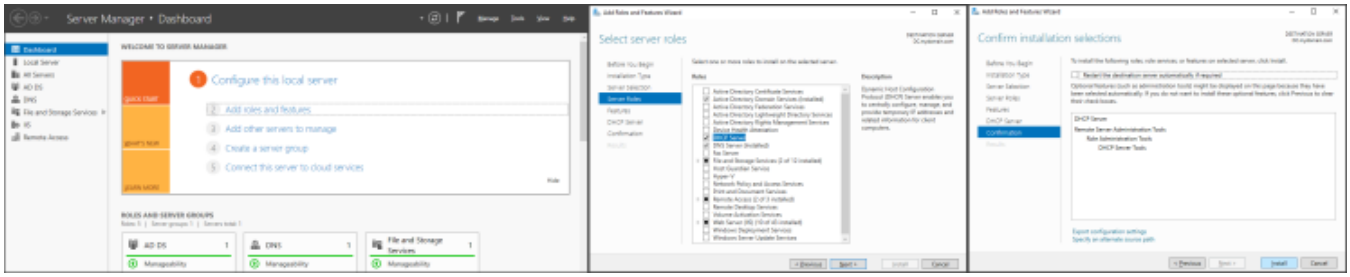
Navigating to Routing and Remote Access, configuring RRAS, and selecting an NAT Internet connection.

Setting Up a DHCP Server

- We will be configuring a **DHCP server** on our domain controller, which will provide IP addresses to our Windows 10 clients. This setup will enable our clients to connect to the internet and browse websites, even though they are connected to a private internal network, similar to an office or school environment.

💡DHCP stands for Dynamic Host Configuration Protocol. It is a network management protocol used to automatically assign IP addresses and other network configuration settings to devices on a network.

- Open **"Server Manager"** and navigate to the **"Add new roles and features"** option. Follow the on-screen instructions, clicking 'Next' until you reach the **"Server Roles"** section. From there, select **"Remote Access"** and click **"Add features"** to include it. Move forward to the **"Confirmation"** section, and initiate the installation by clicking **"Install."**

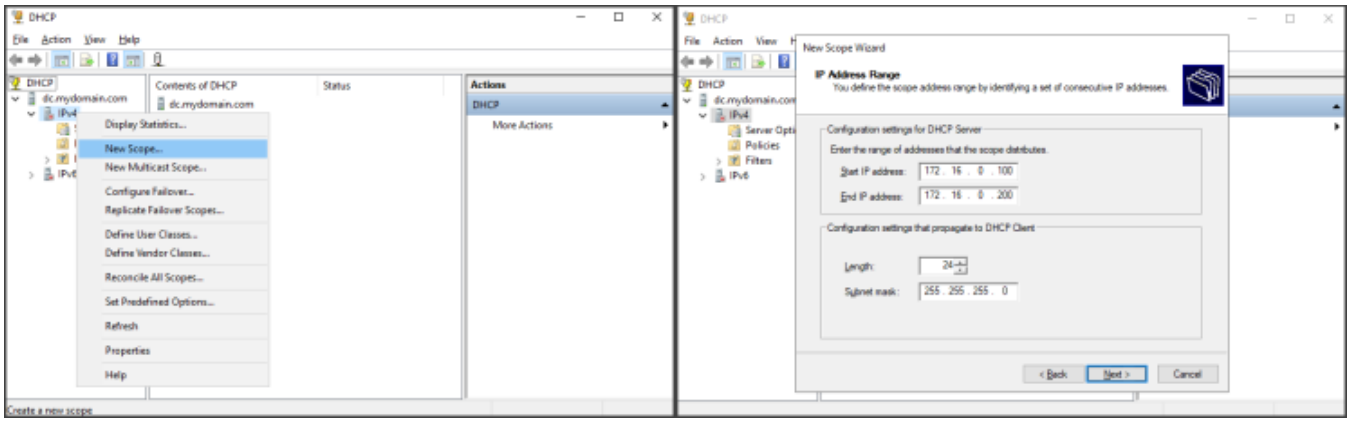


Adding new roles and features, selecting Server Roles and Installing DHCP Server.

- We're going to create a DHCP scope that will assign IP addresses in the range of 172.16.0.100 to 172.16.0.200, using a subnet mask of 255.255.255.0.

- Open "**Server Manager"** and look for the **"Tools"** option at the top-right corner. Click on it, and from the dropdown menu, select **"DHCP"** to access the DHCP Server. Once inside, expand the DHCP Server and right-click on the **"IPv4"** category. In the context menu, choose **"New Scope"** to begin creating a new scope.

- Click "Next" and proceed to the **"Scope Name"** section. Since we are in a lab environment, we will assign the range **"172.16.0.100-200"** to the scope. Click "Next" to continue. For the "IP Address Range" configuration, set the "Start IP address" as 172.16.0.100 and the "End IP address" as "172.16.0.200". Use a "Length" of "24" and a **"Subnet mask"** of **"255.255.255.0"**.



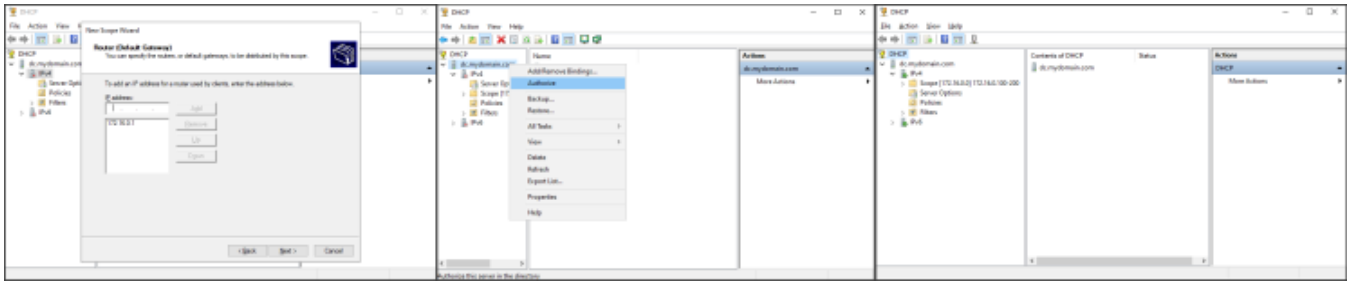
Creating New Scope, and assigning Start and End IP address and Subnet mask.

- Continuing from the previous step, click **"Next"** through the **"Lease Duration"** section without making any changes, as we will leave the preferences as they are for this lab. In the **"Router (Default Gateway)"** section, enter the IP address of your Domain Controller as **"172.16.0.1"** in the **"IP address"** column. Click **"Add"** and then click **"Next"**.

- Moving on to the **"Domain Name and DNS Servers"** section, input **"mydomain.com"** in the **"Parent domain"** column since we are using the domain controller as our DNS server. Click "Next" to proceed. Keep the settings as they are until you reach the end of the wizard, and then click "Finish" to complete the DHCP scope configuration.

- Afterward, right-click on the **DHCP Server** and select **"Authorize".** Then, right-click again and choose "Refresh" to ensure the changes take effect.

💡 *Lease durations determine the period for which devices are assigned IP addresses on a network. They specify how long an IP address is allocated before it needs to be renewed or released.*

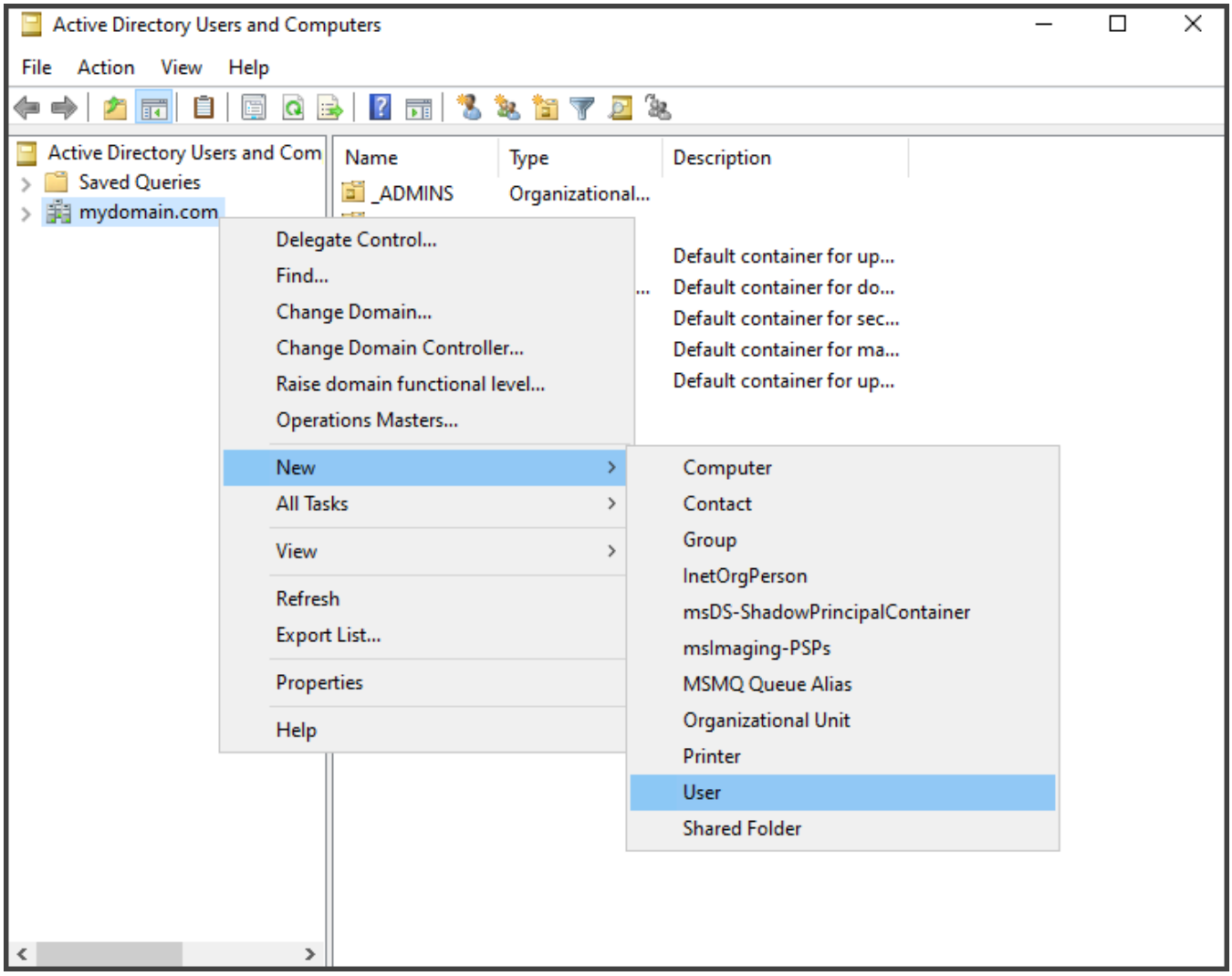


Adding a default gateway, authorizing a new scope in the Domain Controller, and verifying the DHCP server setup.

Adding Domain Users

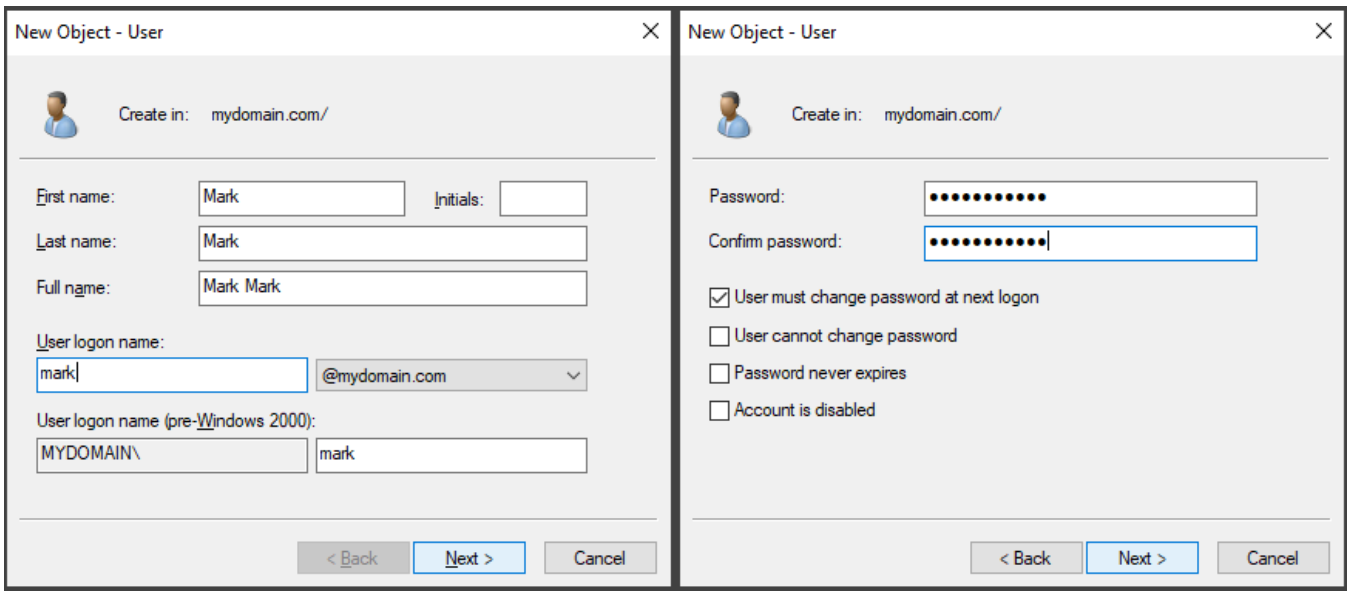
Method #1 Creating Domain Users Manually

- To create a new domain user, access the Server Dashboard, locate the **"Tools"** section, and select **"Active Directory Users and Computers."**
- In the ADUC window, right-click on the domain name **"mydomain.com"** and choose "New" and then "User".



Accessing the ADUC settings and selecting a new User Object.

- Next, provide the necessary user information and set a password for the user. In a typical scenario, it is common to leave the **"User must change password at logon"** option checked. This allows the user to change their password when they log in to their account from their workstation for the first time.To simplify management, for the lab, you can uncheck the box and enable the **"Password never expires"** option.



Creating a new User Object.

Method #2 Automating Domain Users Creation With PowerShell

- The PowerShell script we will utilize, named **"Create Users,"** generates numerous domain users with a default password of **"MS@6264"** and subsequently assigns them to a newly created Organizational Unit named **"_Users."** These domain users are created based on the list of names provided to the script. In the GitHub report, Jon employs a ***names.txt*** file containing a list of 1000 users to generate the domain accounts.

- To suit the scale of our lab, I will customize the script to generate six standard Organizational Units: *Marketing, Finance, Operations, Human Resources, Legal, and IT*. Additionally, the modified script will create **200 domain users.**

- To begin, you can download the zip file from the GitHub repository. Before extracting the downloaded file, it is recommended to disable **Real-Time Monitoring** and set the **PowerShell Execution Policy to "Unrestricted."** I have also customized the script to align with my preferences.

Disable Real-Time Monitoring:

```
Set-MpPreference -DisableRealtimeMonitoring $true -Verbose
```

PowerShell Execution Policy

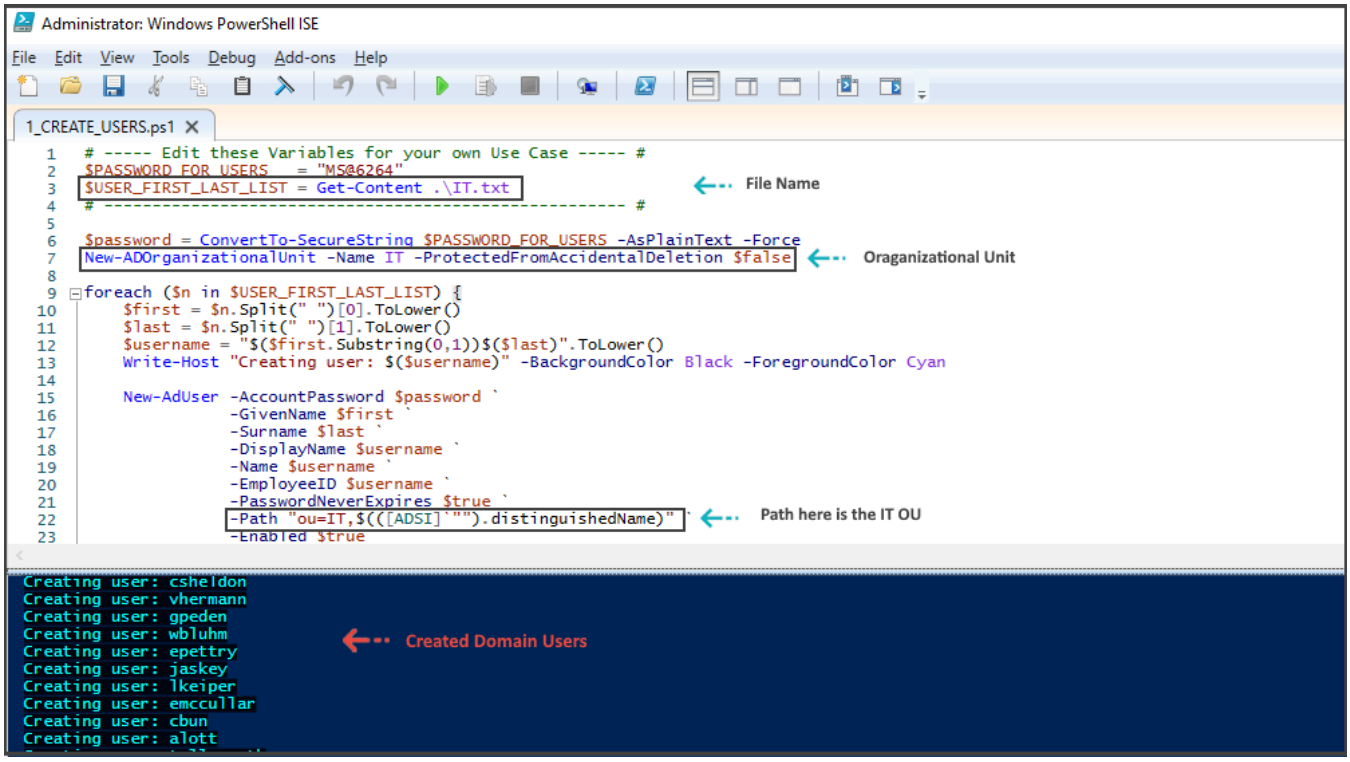
```
Set-ExecutionPolicy Unrestricted
```

```
$PASSWORD_FOR_USERS   = "MS@6264"
$USER_FIRST_LAST_LIST = Get-Content .\IT.txt
# ------------------------------------------------------- #

$password = ConvertTo-SecureString $PASSWORD_FOR_USERS -AsPlair
New-ADOrganizationalUnit -Name IT -ProtectedFromAccidentalDelet

foreach ($n in $USER_FIRST_LAST_LIST) {
    $first = $n.Split(" ")[0].ToLower()
    $last = $n.Split(" ")[1].ToLower()
    $username = "$($first.Substring(0,1))$($last)".ToLower()
    Write-Host "Creating user: $($username)" -BackgroundColor E

    New-AdUser -AccountPassword $password `
               -GivenName $first `
               -Surname $last `
               -DisplayName $username `
               -Name $username `
               -EmployeeID $username `
               -PasswordNeverExpires $true `
               -Path "ou=IT,$(([ADSI]`"").distinguishedName)" `
               -Enabled $true
}
```

- To open the PowerShell script, open it in PowerShell Windows ISE as an **Administrator**. In the script, find the **$USER_FIRST_LAST_LIST** variable and input the file name containing the user's list. If you have divided your 200 domain users into different files based on Organizational Units, make sure to specify the corresponding file for each unit.

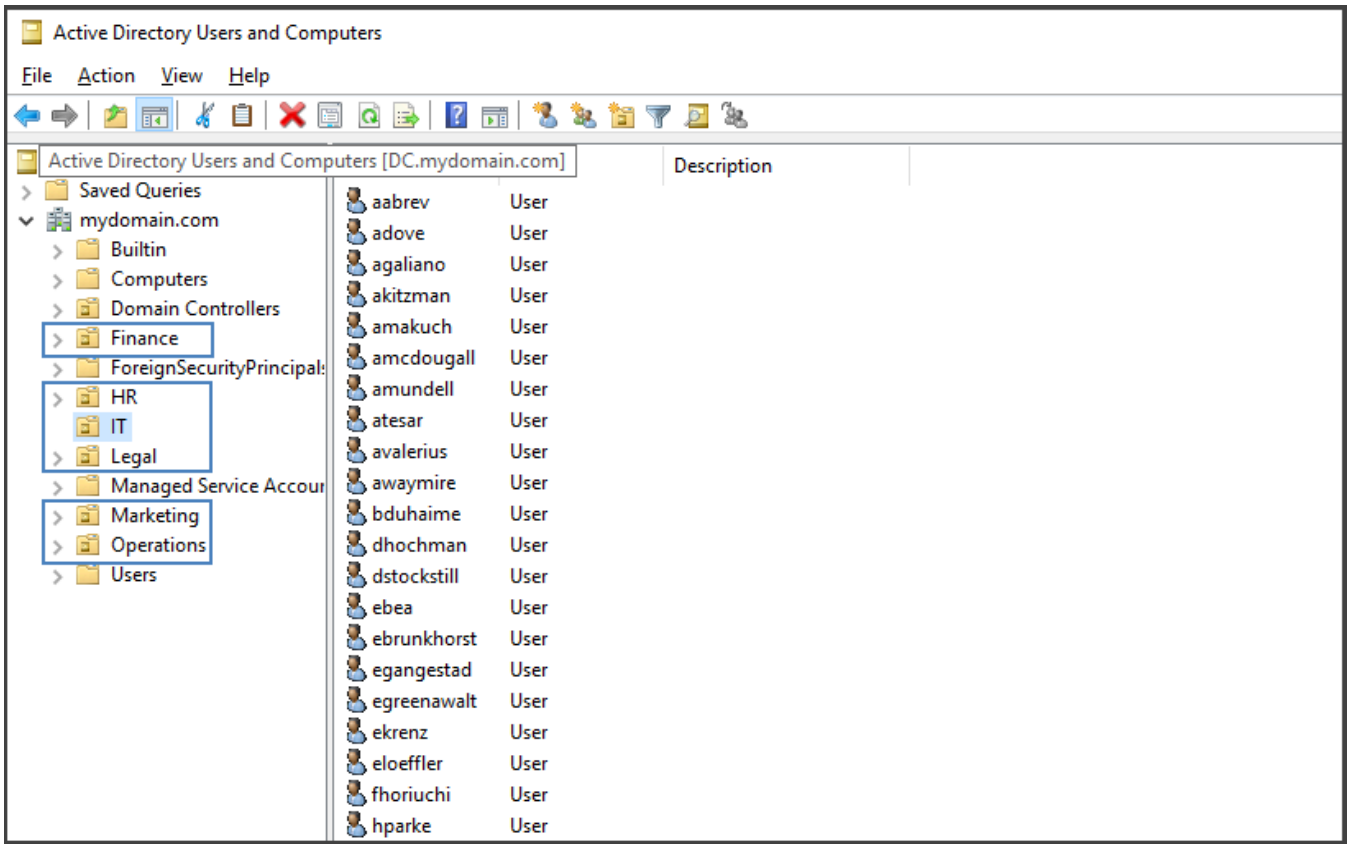
- I began by focusing on the IT department. I assigned the variable **$USER_FIRST_LAST_LIST** to the list of users in that department. Next, I replaced the **New-ADOrganizationalUnit** command, changing the destination from **_Users to IT** to create the OU specifically for the IT department. Additionally, I modified the **-Path** parameter to specify that the users should be added to the OU with the path **ou=IT**.

- After making the necessary changes to the script, simply save it and run it. This will create the domain users and add them to the specified OU as intended.



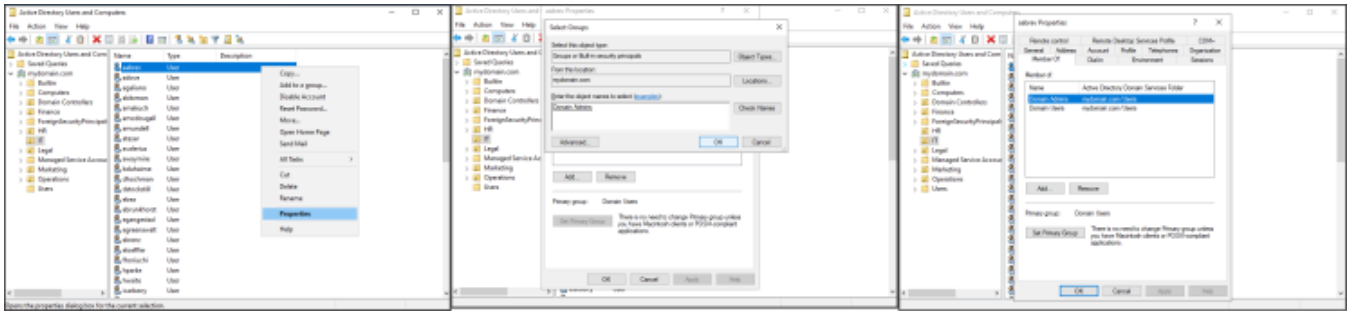
Creating new domain users and add them to the IT OU.

- If you wish to expand your lab setup, you can follow these steps to create additional Organizational Units and add users to them. Upon executing the script, return to the Active Directory Users and Computers window, where you will find the recently generated Organizational Units and domain users.



Newly Creates Organizational Units.

- Given that we now have a sufficient number of users, let's select an individual from the **IT OU** to be designated as the domain admin .
- To assign a User to a group like **“Domain Admins”**, select the desired user like “aabrev”, right-click on their name, and select **“Properties”**.
- To add a user to a specific group, navigate to the **"Member of"** tab. Click on the "Add" button and enter the name of the desired group in the **"Enter the object names"** box. For example, if you want to add the user to the "Domain Admins" group, type "Domain Admins" and click **"Check Names"** to validate the entry. Finally, click "Ok" to finalize the process. Once added, the user's membership in the new group will be displayed in the **"Member of"** list.



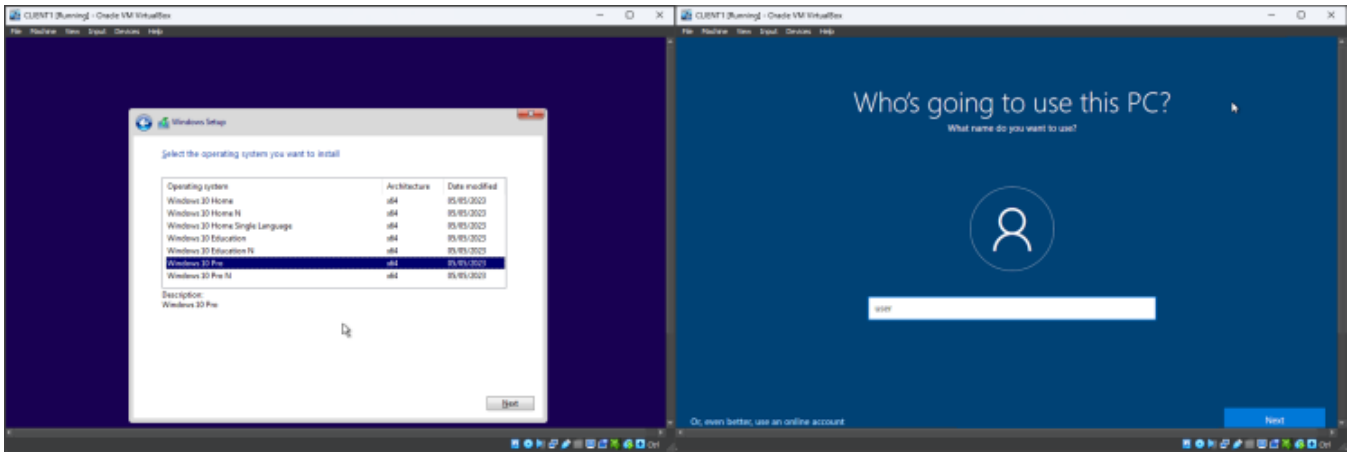
Steps to add a user to a Domain Admin group.

Adding a Client to Active Directory

- We have successfully set up our users, configured the entire environment, and established key components. Our network diagram shows that our **Internet connection, NICs, domain with user setup, NAT, DHCP, and connection to the internal VMware network** are all in place.

- The last step is to create a **Windows 10 virtual machine** in VirtualBox with an internal NIC. It will obtain its IP address from the configured DHCP server. After setting up the virtual machine, we'll confirm that it successfully receives the expected IP address from the DHCP server, ensuring the network setup is functioning properly.

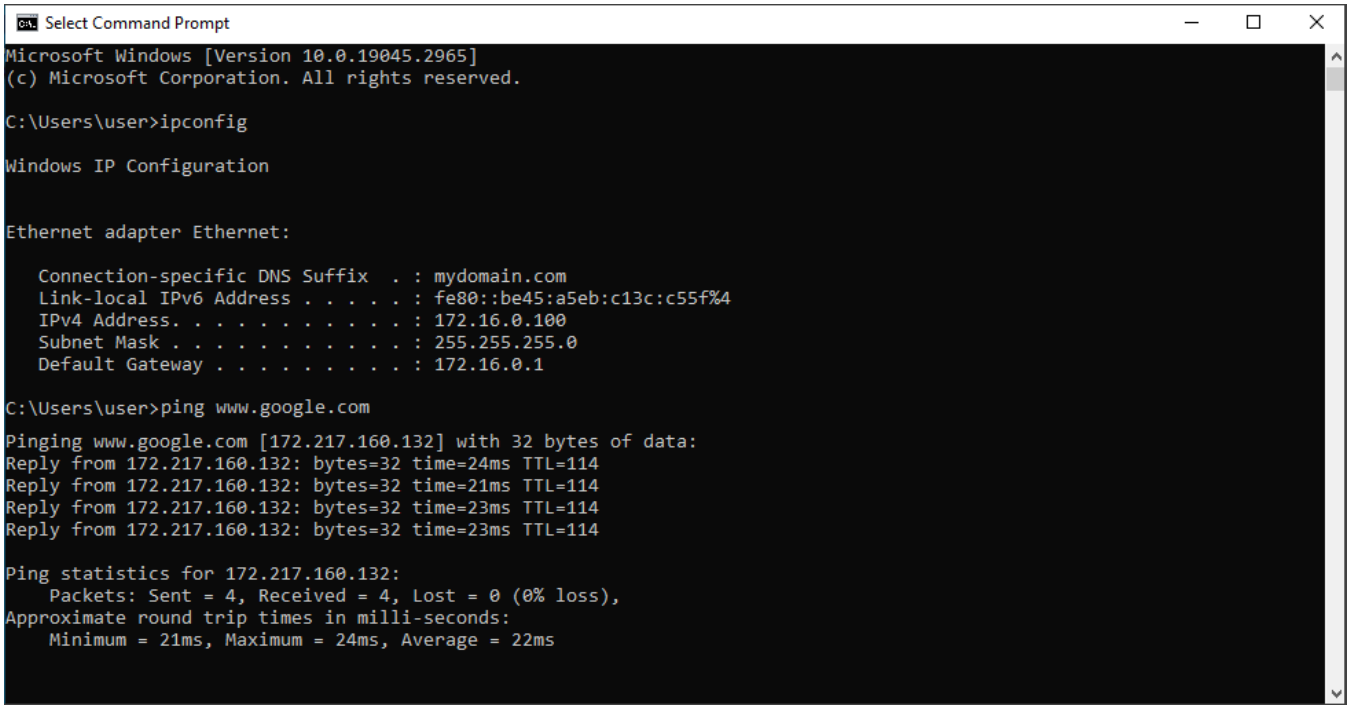
- I followed the same process during the Windows Server 2022 Setup , naming the VM as **"CLIENT1"**. I progressed until I reached the "Select the operating system you want to install" screen in the Windows Setup, where I specifically chose **"Windows 10 Pro"** as the desired operating system. Continuing forward, I arrived at the "Who's going to use this PC?" page and entered "user" as the local username. Since a password was not required, I skipped that section and proceeded to finalize the setup process.



Creating 'CLIENT1' Virtual Machine.

💡*In order to access resources on a Windows Server Domain, it is necessary to have Windows 10 Pro as the Home edition of Windows 10 does not support domain join.*

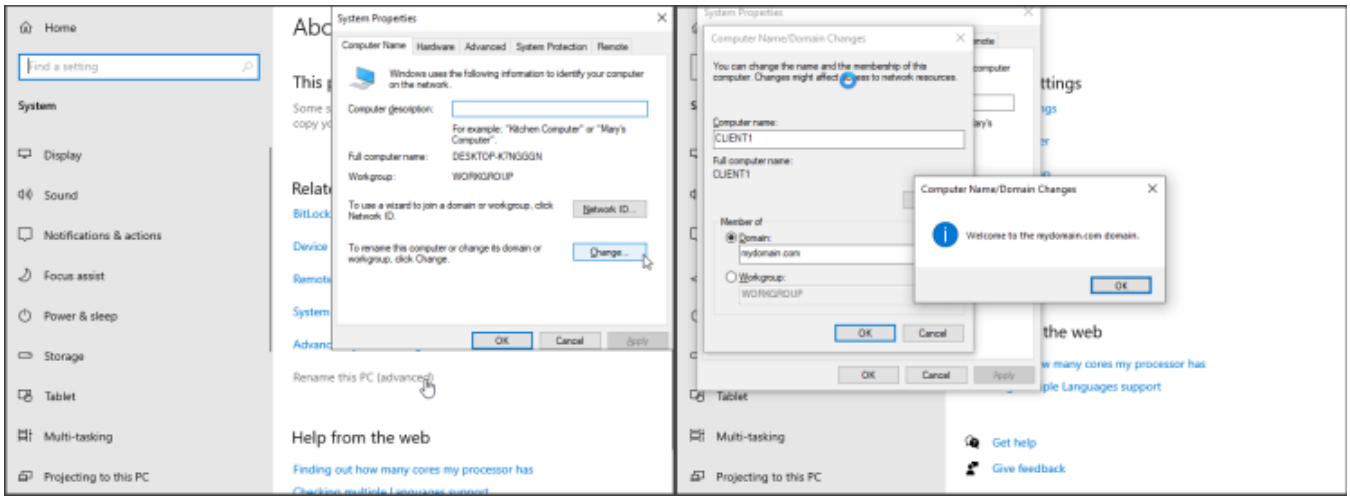
- I confirmed the accurate assignment of the IP address by executing the **"ipconfig"** command in the command prompt. Furthermore, I conducted a successful ping test to google.com, which indicated the operational status of our **DNS server** and established connectivity up to the domain controller, serving as the default gateway. With proper **Network Address Translation (NAT)** in place, the domain controller effectively forwarded the traffic to the internet, enabling the echo reply from the ping to reach us seamlessly. As a result, our network infrastructure, depicted in the network diagram, is functioning as expected.



Verifying Connectivity

- We are going to rename this PC and join the domain, **"right-click"** on the **"Start Menu"** and select **"System."** In the "System" window, locate the option labeled **"Rename this PC (advanced)"** and click on it to open the **"System Properties"** window. Within this window, click on the **"Change"** button.

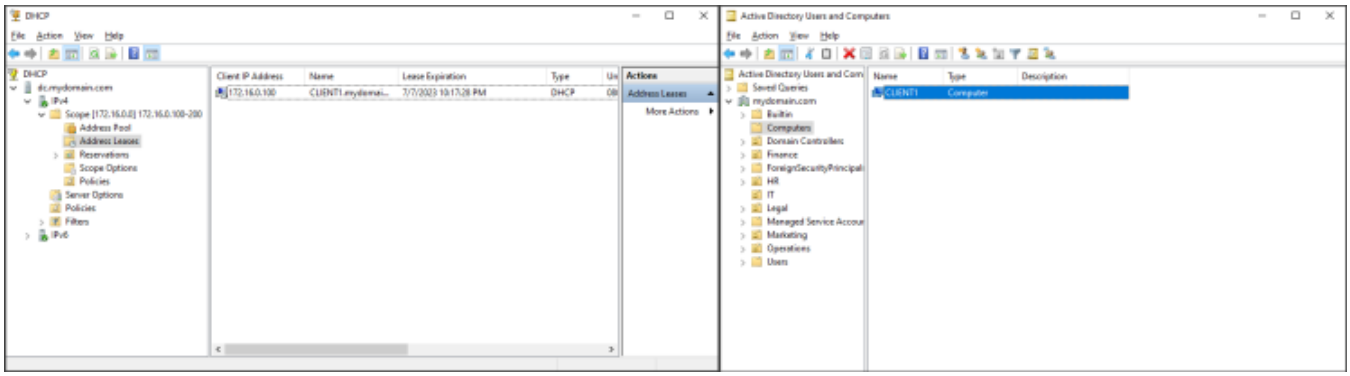
- In the **"Computer Name/Domain Changes"** window, enter **"CLIENT1"** as the new PC name. Choose the **"Domain"** option and enter **"mydomain.com"** as the domain name. Apply the changes by clicking "OK" and provide the necessary administrative credentials to authenticate with the domain controller. Finally, restart the PC by selecting **"Restart Now"** to complete the domain join operation.



Change the computer name and join it to the domain controller.

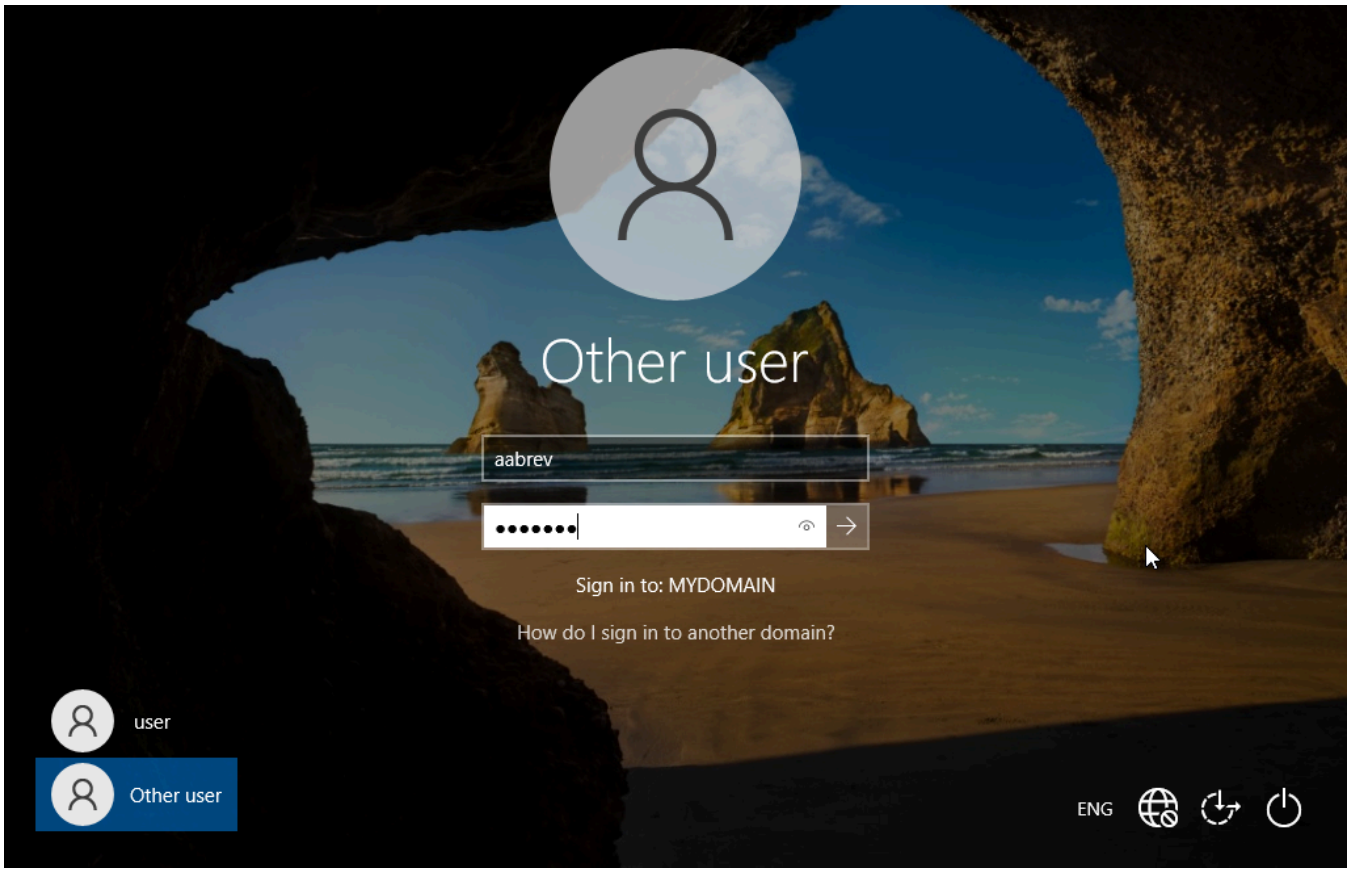
- In the **"Address Leases"** section of the **"DHCP scope"** on the **"Domain Controller"**, we can see that **CLIENT1** is listed. This indicates that when we added the client to the network, it automatically contacted the DHCP server and obtained an IP address assignment.

- Also, In **"Active Directory Users and Computers",** on the "Domain Controller", when we go to the **"Computers"** container, we can see that the client computer automatically appears there after joining the domain, *indicating* its domain membership.



Confirming presence of 'CLIENT1' in DHCP and ADUC.

- Since the client computer has joined the domain, we can use any of the created accounts to log in. Our next step is to proceed with logging into the computer using the username **"aabrev"** and password **"MS@6264".**



Logging in using a generated user account.

- We can observe that we are successfully logged in, and our desktop is functioning as expected. By accessing the command line and executing the **"who am i"** command, we can verify our domain membership and authenticate our username as **"aabrev."**

Verifying username using command promt.

In conclusion, setting up an Active Directory environment in a home lab using Oracle VirtualBox offers invaluable opportunities for learning and skill development in network administration and Windows server technologies. Throughout this guide, I've meticulously walked through each step, from

configuring the domain controller to integrating client machines and automating user creation with PowerShell scripting.

By following these instructions and engaging in hands-on practice, one can gain a deep understanding of Active Directory concepts, DNS, DHCP, NAT, and other essential services. The visual aids provided, including network diagrams and screenshots, enhance comprehension and facilitate smooth implementation.

Happy labbing!

Thank you for taking the time to read this article. Your feedback and suggestions are greatly appreciated!

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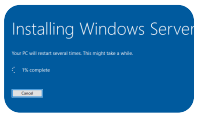

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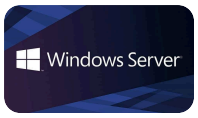

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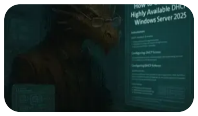

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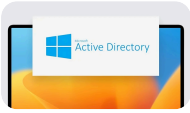

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